Displacement and velocity

1 The position (in metres) of an object along a straight line after t seconds is modelled by:

$$x(t) = 6t^2$$

a State the velocity $v(t)$ of the object at time t.

b State whether the following represent the velocity of the object after 4 seconds:

i $x'(4)$ **ii** $v'(4)$ **iii** $x(4)$ **iv** $v(4)$

c Hence, find the velocity of the object after 4 seconds.

2 The position (in metres) of an object along a straight line after t seconds is modelled by:

$$x(t) = 3t^3 - 4t^2$$

a State the velocity $v(t)$ of the object at time t.

b Hence, find the velocity of the object after 2 seconds.

3 The position (in metres) of an object along a straight line after t seconds is modelled by:

$$x(t) = 3t^2 + 5t + 2$$

a Find $v(t)$, the velocity function.

b Find the velocity of the object after 4 seconds.

4 The position (in metres) of an object along a straight line after t seconds is modelled by:

$$x(t) = 18\sqrt{t}$$

a Find $v(t)$, the velocity function.

b Find the velocity of the object after 9 seconds.

5 The position (in metres) of a drag racer from the starting line at a given time t seconds can be modelled by the equation:

$$x(t) = \frac{t^3}{2} + 3t^2$$

a The drag car finishes the race in 9 seconds. Find its average speed in $\mathrm{m/s}$.

b Find $v(t)$, the velocity function.

c Find the velocity of the car after 5 seconds.

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d At what time does the car have a velocity of $18\ \mathrm{m/s}$?

6 Sophia is throwing a ball straight up in the air. The height, h, above the point where the ball is released is given by the equation $h = -4.9x^2 + 11.27x$.

a Sophia catches the ball after 2.3 seconds. Find the average upwards velocity over the time the ball was in the air.

b Find $v\,(t)$, the velocity function.

c Find the upward velocity of the ball when first released.

d Find the upward velocity of the ball after 0.575 seconds.

e Find the upward velocity after 1.15 seconds.

7 The height in metres of a projectile above level, flat ground is given by $h = 9 + 8t - t^2$, where t is given in seconds.

a State the initial height of the projectile.

b Find the maximum height reached.

c At what time will the projectile reach the ground?

d Sketch the graph of the height of the projectile over time.

e How far will the projectile travel in the first 5 seconds?

8 A boy stands on the edge of a sea-cliff with a height of $48\ \mathrm{m}$. He throws a stone off the cliff so that its vertical height above the cliff is given by $h = 16t - 4t^2$ where t is given in seconds.

a Find the maximum value of h reached by the stone above the cliff top.

b Find the time that elapses before the stone hits the ocean below.

c Find the velocity with which the stone hits the ocean.

9 The displacement (in metres) of a particle moving in rectilinear motion after t seconds is modelled by $x\,(t) = 2t^3 - 7t^2 + 3t - 4$.

a Find the initial displacement of the particle.

b Find $v\,(t)$, the velocity function.

c Find the initial velocity of the particle.

d Find the speed of the particle when $t = 2$.

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10 The displacement (in metres) of a particle moving in rectilinear motion after t seconds is modelled by:

$$x(t) = (4t + 3)^2$$

a Find the initial displacement of the particle.

b Find $v(t)$, the velocity function.

c Find the initial velocity of the particle.

d Find the initial speed of the particle.

11 The displacement (in metres) of a particle moving in rectilinear motion after t seconds is modelled by:

$$x(t) = -5t(t - 4)$$

a Find the initial displacement of the particle.

b Find the time t when the particle next returns to the origin.

c Find the distance travelled by the particle between leaving the origin and returning again.

12 The displacement (in metres) of a particle moving in rectilinear motion after t seconds is modelled by:

$$x(t) = -2t(t + 10)$$

a Find the initial displacement of the particle.

b Find $v(t)$, the velocity function.

c Find the initial velocity of the particle.

d Find the initial speed of the particle.

13 Let $s = 10 + 12t - 4.7t^2$ be the height of an object in metres at time t in seconds.

a Find the velocity of the object in terms of t.

b Find the acceleration of the object in terms of t.

14 A boy throws a ball vertically. The height of the ball h in metres is given by $h = 1 + 17t - 5t^2$ where t is given in seconds.

a Find the velocity of the ball after 3 seconds.

b Find the acceleration due to gravity.

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15 The displacement (in metres) of a body moving along a straight line after t seconds is modelled by:

$$x(t) = -t^3 + at^2 + bt + 7$$

The initial velocity of the body is $8\ \mathrm{m/s}$. The body is momentarily at rest at $t = 2$ seconds.

Find the values of a and b.

Acceleration

16 The displacement (in metres) of a particle moving in rectilinear motion after t seconds is modelled by:

$$x(t) = 2t^2 - 4t + 5$$

- **a** Find $v(t)$, the velocity function.
- **b** Find the time t at which the particle changes direction.
- **c** Find $a(t)$, the acceleration function.

17 The displacement (in metres) of a body from an origin O at time t seconds is modelled by:

$$x(t) = t^2 - 7t + 5$$

- **a** Find the velocity function, $v(t)$.
- **b** Find the initial velocity of the body.
- **c** Find the acceleration function, $a(t)$.
- **d** Find the acceleration of the body at $t = 8$.
- **e** Find the value(s) of t for which the body has a velocity of $3\ \mathrm{m/s}$.
- **f** Find the value(s) of t for which the body has a speed of $3\ \mathrm{m/s}$.

18 A car starts at rest and has a displacement of s m in t seconds, where $s = \frac{1}{6}t^3 + \frac{1}{4}t^2$.

- **a** Find the acceleration of the car in terms of t.
- **b** Find the initial acceleration of the car.
- **c** Find the acceleration of the car when $t = 16$.

19 A car starts at rest and has a displacement of s m in t seconds, where $s = t^4 + t^2$.

- **a** Find the acceleration of the car in terms of t.
- **b** Find the acceleration of the car when $t = 0$.

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c Find the acceleration of the car when $t = 3$.

20 A particle moves in a straight line and its displacement after t seconds is given by:

$$x = 19t^2 - 2t^3$$

where x is its displacement in metres.

Let v and a represent its velocity and acceleration at time t respectively.

a Find an expression for the acceleration a of the particle after t seconds.

b Find the acceleration of the particle after 4 seconds.

c State whether the particle is speeding up or slowing down after 4 seconds.

21 A particle travels in a straight line such that its displacement, x, in metres from a fixed point is given by:

$$x = 12 + 6t^2 - t^3$$

where t is the time in seconds after it starts moving.

a Find the time $t > 0$ at which the particle instantaneously comes to rest.

b State the velocity of the particle at this time when it comes to rest.

c Find the time t at which the acceleration of the particle is zero.

d Find the velocity of the particle when its acceleration is 0.

e When the acceleration of the particle is 0, what does this imply about the velocity?

22 A particle moves in a straight line, starting from rest at the point O. At time t seconds after leaving O, the speed v of the particle (in $\mathrm{m/s}$) is given by:

$$v(t) = \left(3t - 2t^2\right)^2$$

a Find the time $t > 0$ at which the particle instantaneously comes to rest.

b Determine $a(2)$, the acceleration of the particle after 2 seconds.

c Describe the motion of the particle.

23 A particle moves in a straight line. Its velocity (in metres per second), t seconds after passing the origin is given by:

$$v = 2t^2 - 10t$$

Find the velocity when the acceleration of the particle is zero.

24 The position of an object moving along the x-axis at time t seconds is given by:

$$x = t^3 - 6t^2 + 3t - 2$$

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Its velocity and acceleration at time t seconds are given by v and a respectively. Find the position x of the object when its acceleration is 0.

25 The velocity (in metres per second) of a body moving in rectilinear motion after t seconds is modelled by:

$$v(t) = t^2 - 11t + 24$$

a Find the time(s), t, when the body is instantaneously at rest.

b Find $a(t)$, the acceleration function.

c Find the acceleration at time $t = 2$.

26 The velocity (in metres per second) of a body moving in rectilinear motion after t seconds is modelled by:

$$v(t) = 5t^2 - 23t + 24$$

a Find the time(s), t, when the body is instantaneously at rest.

b Find $a(t)$, the acceleration function.

c Find the acceleration at time $t = 3$.

27 An object, P, moves in rectilinear motion such that its displacement x (in metres) from the origin at time t seconds is given by:

$$x(t) = (t + 1)(4 - t)(t + 4)$$

Its velocity and acceleration at time t are given by $x'(t)$ and $x''(t)$ respectively.

a Find the time t at which the particle reverses direction.

b Find the equation for the acceleration of the particle at time t.

c Find the acceleration when the particle is instantaneously at rest.

d Find the average velocity of the particle in the first 3 seconds.

28 The distance d in metres travelled by a train between two stations is given by:

$$d = 0.86t^2 - 0.0084t^3$$

where t is given in seconds.

a Write an equation for the velocity of the train in metres per second.

b Find the time taken to travel between the two stations. Round your answer to the nearest second.

c Find the distance between the two train stations. Round your answer to the nearest metre.

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d Find the maximum velocity of the train. Round your answer to the nearest m/s.

e Find the maximum acceleration of the train.

29 The following table represents data for a typical driver approaching a red traffic light. At a given speed, the driver must first react and then press the brake. The distance travelled during this period is called the thinking distance. After the brakes are applied, the driver slows down to a stop and covers a distance called the braking distance.

Speed (km/h)	Thinking distance (m)	Breaking distance (m)
36	9	9
48	12	16
60	15	25
72	18	36
84	21	49
96	24	64

a Find the thinking time for a driver travelling at $96\ \mathrm{km/h}$.

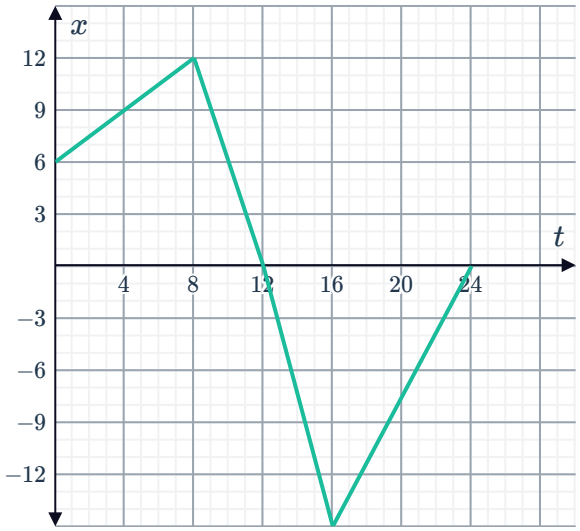
b The distance $s\ (t)$ travelled by an object travelling at an initial speed of u m/s, braking with acceleration $a\ \mathrm{m/s^2}$ is given by $s\ (t) = ut + \frac{1}{2}at^2$. Find the derivative of s.

c Hence find t when a car with initial velocity, u , comes to rest.

d Find the braking acceleration (in $\mathrm{m/s^2}$) for the driver travelling at $96\ \mathrm{km/h}$.

Graphs of displacement, velocity and acceleration

30 A ball moves horizontally in a $30\ \mathrm{m}$ long tube. Its position from the centre is shown by the following graph:

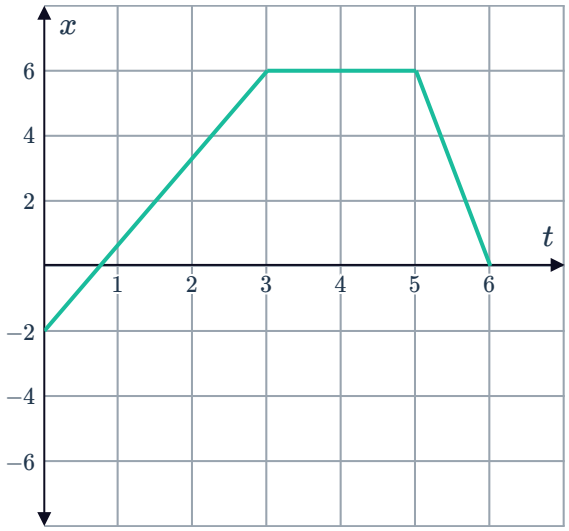


a State the difference in displacement of the ball between $12\ \mathrm{s}$ and $24\ \mathrm{s}$.

b Find the total distance travelled by the ball between $12\ \mathrm{s}$ and $24\ \mathrm{s}$.

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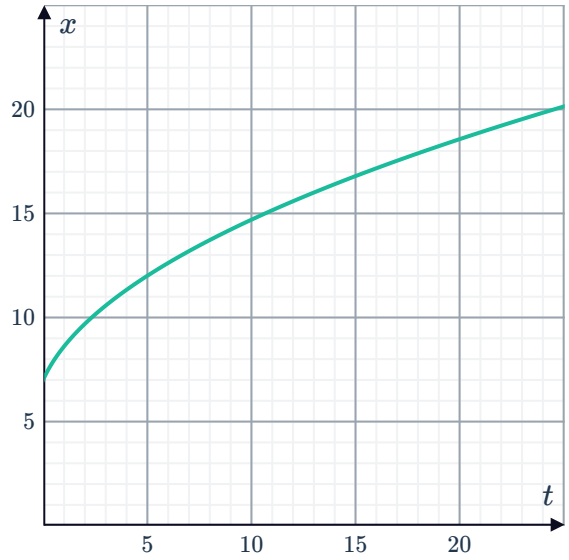
31 A ball moves horizontally along a straight line. Its position from the centre is shown by the following graph:



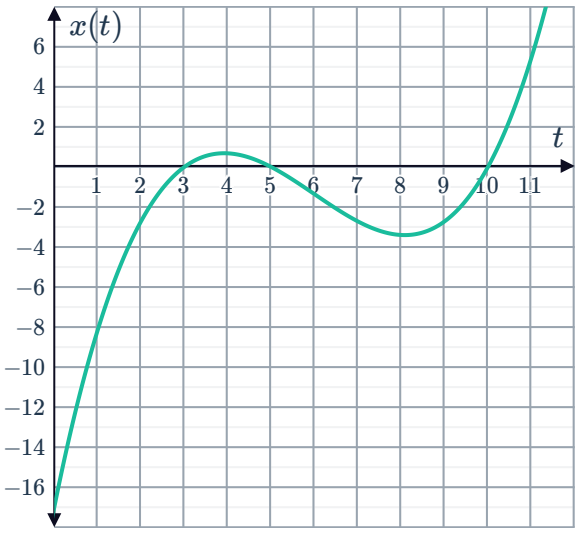
a State the difference in displacement of the ball between 3 s and 6 s.

b Find the distance travelled by the ball between 3 s and 6 s.

32 The graph shows the particle's position against time. Sketch the graph of the particle's velocity against time.



33 The position of a particle relative to a fixed position O along a straight line is shown by the following graph. The particle starts -17 m to the left of the point O.



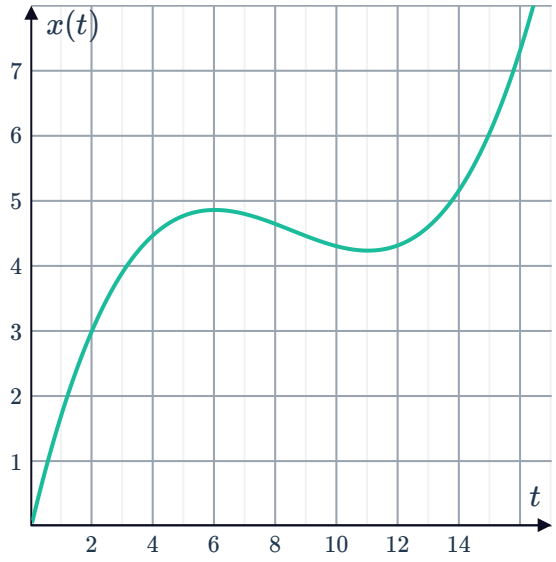
a State the times at which the particle is at the point O.

b For what values of t is the velocity positive?

c For what values of t is the velocity equal to zero?

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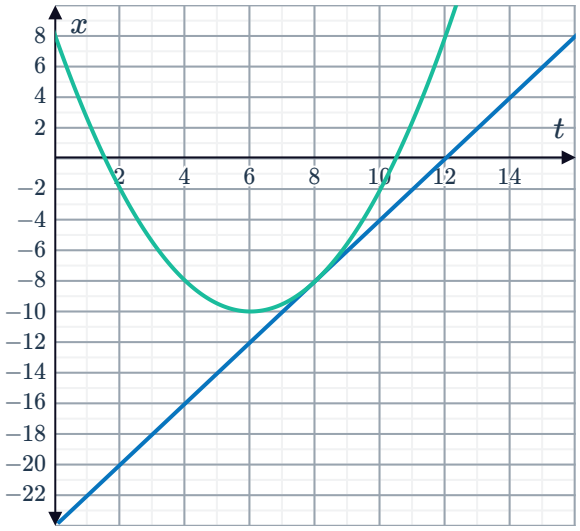
34 The position of a particle relative to a fixed position O along a straight line is shown by the following graph. If the particle is positioned to the right of the point O, then it has positive displacement.



a At which times is the velocity zero?

b For what values of t is the velocity negative?

35 The position of a particle relative to a fixed position O along a straight line is shown by the following graph, along with a tangent line at $t = 8$. If the particle is positioned to the right of the point O, then it has positive displacement.



a State the position of the particle when:

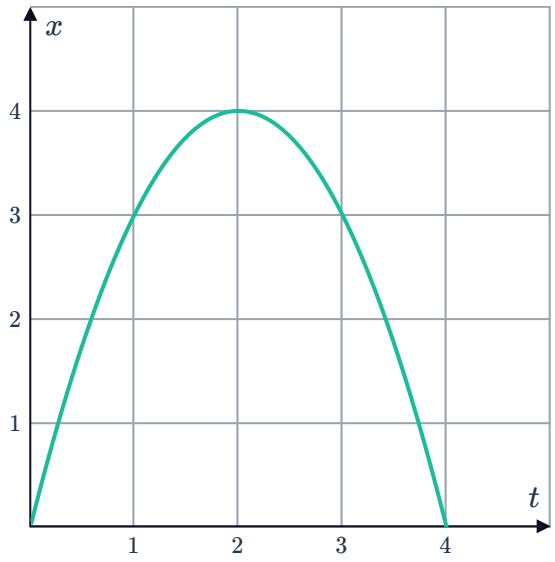
i $t = 2$ **ii** $t = 12$

b Find the average velocity of the particle from $t = 2$ and $t = 4$.

c Find the instantaneous velocity at $t = 8$.

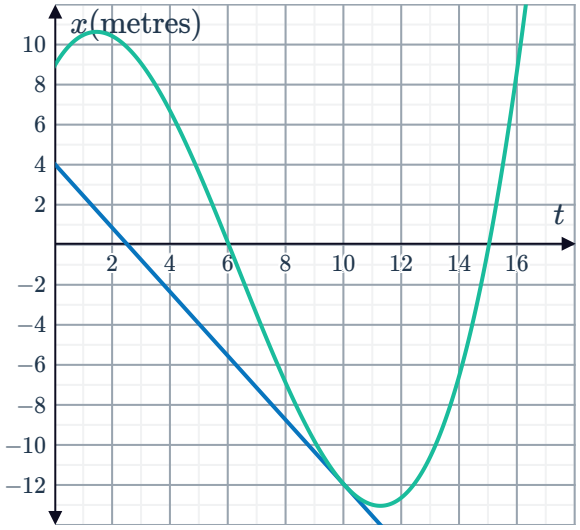
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36 The position of a particle relative to a fixed position O along a straight line is shown by the following graph. If the particle is positioned to the right of the point O, then it has positive displacement.



- **a** State the value of t where the particle is stationary.
- **b** State the values of t where the particle is moving to the right.
- **c** How far from the point O does the particle go to the right?
- **d** Find an expression for x in terms of t.
- **e** How long does it take to return to the point O?
- **f** Find the velocity of the particle in terms of t.
- **g** Find the velocity when $t = 3$.

37 The position of a particle relative to a fixed position O along a straight line is shown by the following graph, along with a tangent line at $t = 10$. If the particle is positioned to the right of the point O, then it has positive displacement.



- **a** State the initial position of the particle.
- **b** At what times does the particle return to the point O?
- **c** In which direction is the particle moving at $t = 5$?
- **d** At what time does the particle change from moving left to right?
- **e** Find the instantaneous velocity of the particle when $t = 10$.

38 An object moves in a straight line, with its position from a point O given by the equation:

$$x(t) = t^3 - 4t^2 + 5$$

- **a** Find $v(t)$, the velocity function.
- **b** Find the exact times at which the object changes direction.
- **c** Sketch a graph of the object's position over time.
- **d** Find the total distance that the object has travelled by the time $t = \frac{8}{3}$ units.

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39 A particle, P, starts from rest at point O, and its velocity t seconds after it starts moving is modelled by the equation:

$$v = t^2 (9 - t)$$

After 9 seconds it comes to rest and stops moving.

a Find an equation for the acceleration a of the particle after t seconds.

b At what time $0 < t < 9$ is the velocity of the particle greatest?

c Find the greatest velocity that the particle obtains in the first 9 seconds.

d Sketch the graph of the acceleration of the particle over the first 9 seconds on a number plane.

e State the magnitude of the greatest acceleration that the particle achieves in the first 9 seconds.

40 A particle moves in a straight line and its displacement after t seconds is given by:

$$x = 12t - 2t^2$$

where x is its displacement in metres from the starting point. Let v and a represent its velocity and acceleration at time t respectively.

a Find an equation for the velocity, v, of the particle after t seconds.

b After how many seconds, t, does the particle change its direction of motion?

c Sketch the graph of displacement against time.

d Find the displacement of the particle after 9 seconds.

e Hence find the total distance that the particle has travelled in the first 9 seconds.

41 The position of a particle in metres relative to a point O along a straight line is given by:

$$x(t) = t^3 - 9t^2 + 24t$$

where t is in seconds.

a Find the velocity of the particle in terms of t.

b Find the acceleration of the particle in terms of t.

c Construct a sign diagram for the particle's:

i Velocity **ii** Acceleration

d State the values of t where the particle changes direction.

e At what times is the particle's speed decreasing?

f At what times is the particle's velocity decreasing?

g Find the total distance travelled by the particle in the first 5 seconds.

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42 A particle passes through a hole in a wall. It's displacement from the wall, s metres, at time t seconds after it passes through the hole is given by:

$$s = \frac{1}{3}t + \frac{1}{9}t^2 - \frac{1}{27}t^3$$

In this case, t may be negative.

a Sketch the graph of s for $t \in [-3, 6]$.

b Find the velocity of the particle in terms of t.

c Sketch the graph of $\frac{ds}{dt}$ for $t \in [-3, 6]$.

d The wall is positioned between two points A and B, at which the particle stops and changes direction. How long does it take for the particle to travel from point A before passing through the wall, to point B after passing through the wall?

e Find the displacement from the wall at point A.

f Find the displacement from the wall at point B.

g Find the average velocity between the two points.

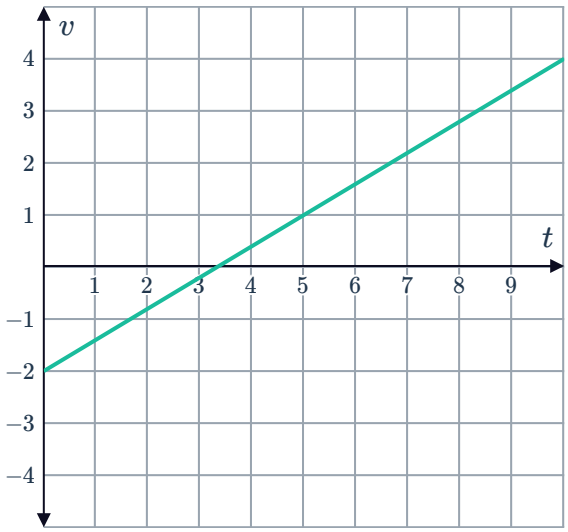
Displacement and acceleration from velocity

43 Consider each of the following velocity graphs for a particular particle where t is in seconds and $t \in [0, 10]$. Find the values of t for which the particle is:

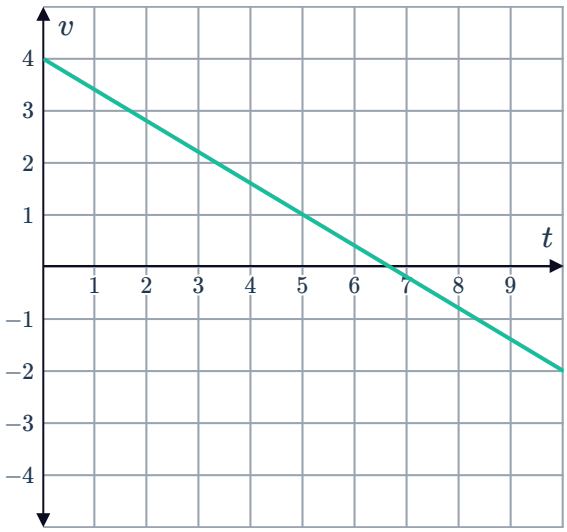
i Accelerating

ii Decelerating

a

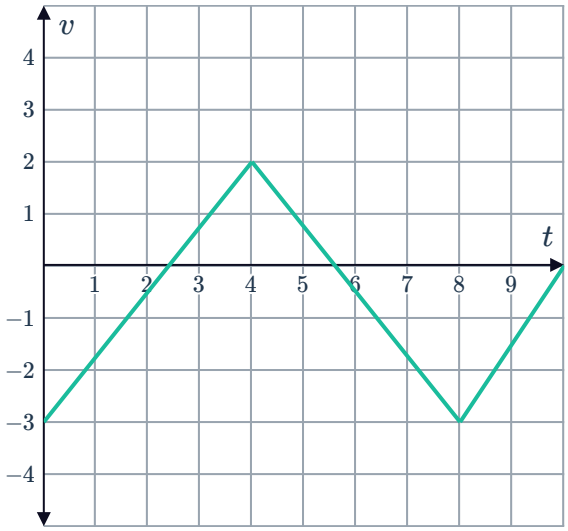


b

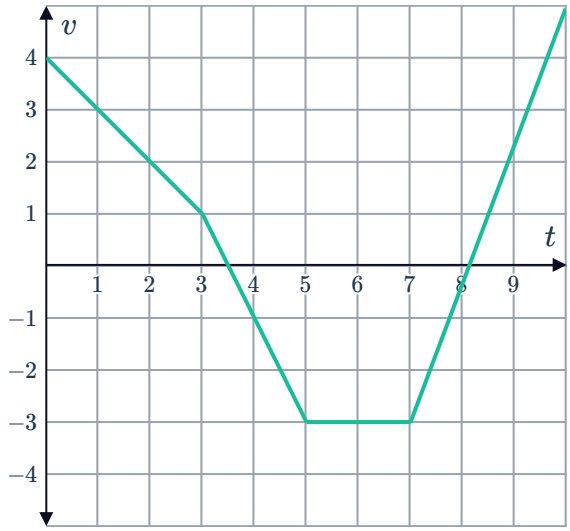


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c



d

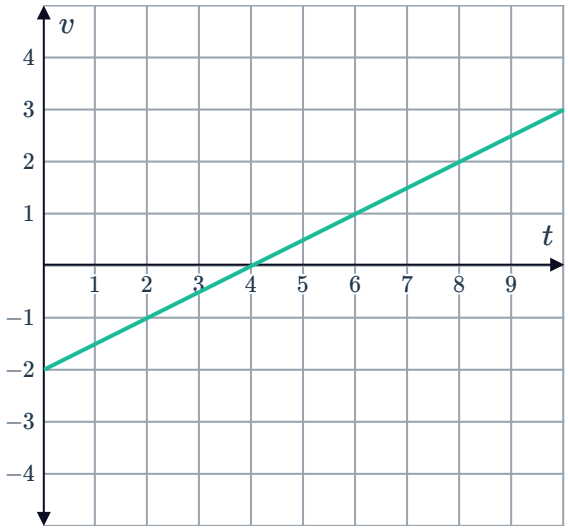


44 Consider the following velocity graphs for a particular particle where t is in seconds.

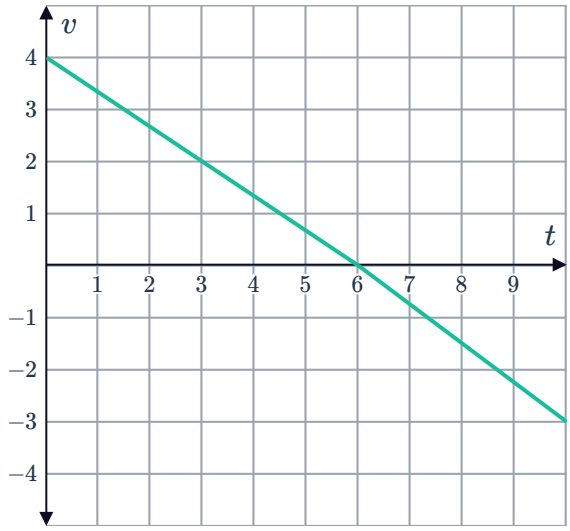
i Find the displacement of the particle after the first ten seconds.

ii Find the total distance travelled by the particle in the first ten seconds.

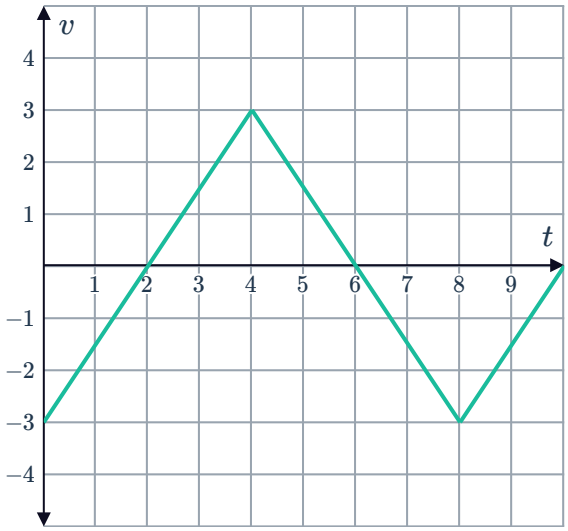
a



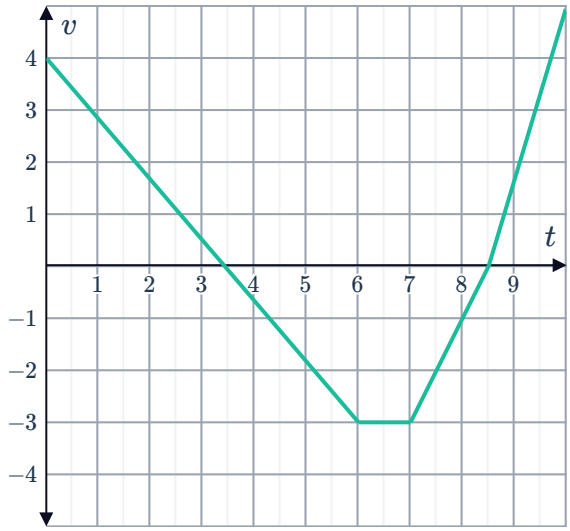
b



c



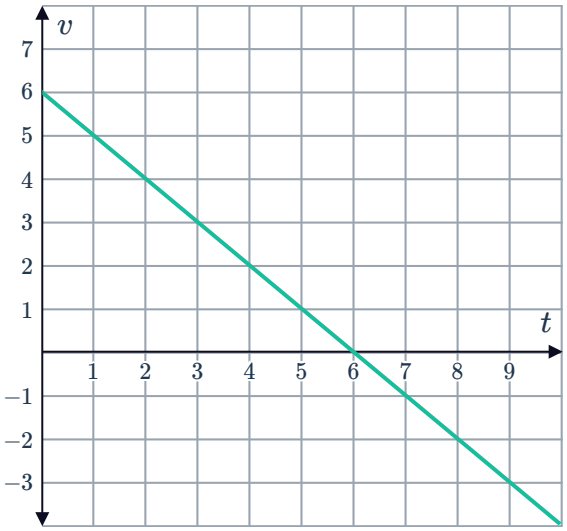
d



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45 Consider the following velocity function for a particular particle where t is in seconds:

Explain why the displacement of the particle after the first six seconds is equal to the total distance travelled in the first six seconds.



46 A particle moves such that its velocity is given by $v(t) = 2t - 8$.

a Sketch the graph of the function of v against t for $0 \leq t \leq 5$.

b Find the displacement of the particle after the first five seconds.

c Find the total distance travelled by the particle in the first five seconds.

47 A particle moves such that its velocity is given by $v(t) = 12 - 4t$.

a Sketch the graph of the function of v against t for $0 \leq t \leq 8$.

b Find the displacement of the particle from $t = 1$ seconds to $t = 6$ seconds.

c Find the total distance travelled by the particle from $t = 1$ seconds to $t = 6$ seconds.

48 A particle moves such that its velocity is given by $v(t) = \dfrac{t}{2} - 4$.

a Sketch the graph of the function of v against t for $0 \leq t \leq 12$.

b Find the displacement of the particle from $t = 4$ seconds to $t = 10$ seconds.

c Find the total distance travelled from $t = 4$ seconds to $t = 10$ seconds.

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